Multiple Observations in Same Subject

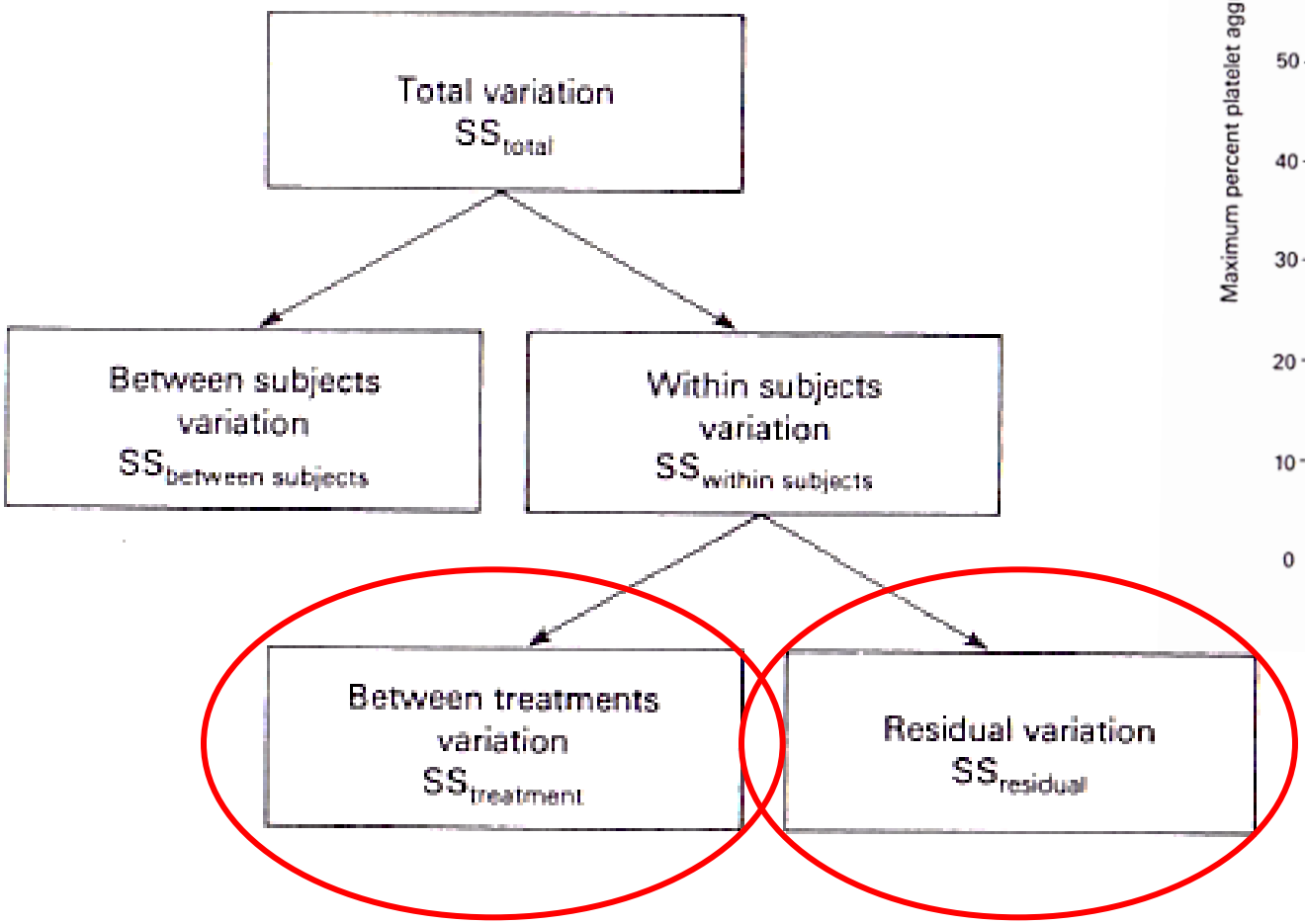
"Repeated Measures"

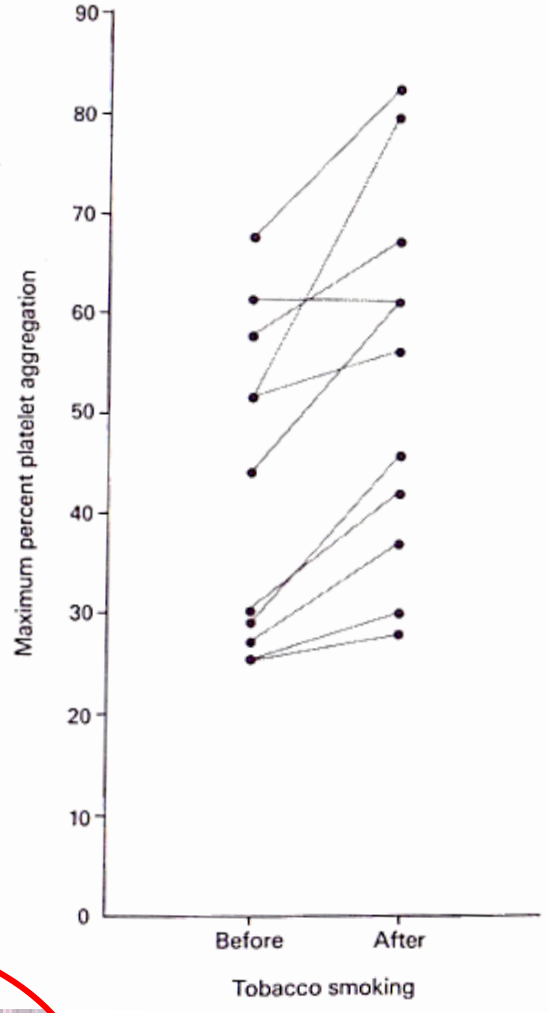
II. Repeated Measures ANOVA
  A. Experiment Framework
  B. Partitioning Variability
  C. Procedure
  D. Examples

III. Isolating Differences in Repeated Measures ANOVA

IV. Repeated Measures w/ Nominal Data – McNemar's Test

II. Repeated Measures ANOVA





$$F = \frac{MS_{treat}}{MS_{res}}$$

II. Repeated Measures ANOVA
A. Framework

Table 9-4 Notation for Repeated-Measures Analysis of Variance

Experimental subject, $n = 4$	Treatment, $m = 3$			Subject	
	1	2	3	Mean	SS
1	X_{11}	X_{21}	X_{31}	$\overline{S}_1$	$\sum_t (X_{t1} - \overline{S}_1)^2$
2	X_{12}	X_{22}	X_{32}	$\overline{S}_2$	$\sum_t (X_{t2} - \overline{S}_2)^2$
3	X_{13}	X_{23}	X_{33}	$\overline{S}_3$	$\sum_t (X_{t3} - \overline{S}_3)^2$
4	X_{14}	X_{24}	X_{34}	$\overline{S}_4$	$\sum_t (X_{t4} - \overline{S}_4)^2$
Mean	$\overline{T}_1$	$\overline{T}_2$	$\overline{T}_3$		

$$\text{Grand mean } \overline{X} = \frac{\sum_t \sum_s X_{ts}}{mn} \qquad SS_{tot} = \sum_t \sum_s (X_{ts} - \overline{X})^2$$

x_{ts} t – treatment group $t = 1, \ldots, m$

s – subject number $s = 1, \ldots, n$

1. Subject Means: $\bar{S}_1, \bar{S}_2, \ldots, \bar{S}_n$ $\left(\bar{S}_s = \frac{1}{m} \sum_{t=1}^{m} x_{ts} \right)$

2. Treatment Means: $\bar{T}_1, \bar{T}_2, \ldots, \bar{T}_m$ $\left(\bar{T}_t = \frac{1}{n} \sum_{s=1}^{n} x_{ts} \right)$

3. Grand Mean: $\bar{X} = \frac{1}{mn} \sum_{t=1}^{m} \sum_{s=1}^{n} x_{ts} = \frac{1}{m} \sum_{t=1}^{m} \bar{T}_t = \frac{1}{n} \sum_{s=1}^{n} \bar{S}_s$

4. Total Sum of Squares: $SS_{tot} = \sum_{t=1}^{m} \sum_{s=1}^{n} \left(x_{ts} - \bar{X} \right)^2$ $DF_{tot} = mn - 1$

Individual measurement

Grand mean

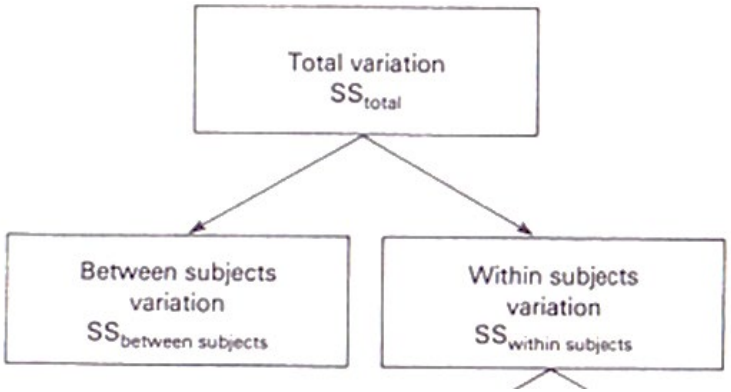
x_{ts} t – treatment group $t = 1, \ldots, m$

s – subject number $s = 1, \ldots, n$

5. Sum of Squares Within Subjects:

$$SS_{wit\,subjs} = \sum_{s=1}^{n} \underbrace{\left[\sum_{t=1}^{m} \left(x_{ts} - \bar{S}_s \right)^2 \right]}_{SS_{subj}} = \sum_{s=1}^{n} SS_s$$

$$DF_{wit\,subjs} = n(m-1)$$



6. Sum of Squares Between Subjects:

$$SS_{bet\,subjs} = m \sum_{s=1}^{n} \left(\bar{S}_s - \bar{X} \right)^2$$

$$DF_{bet\,subjs} = n - 1$$

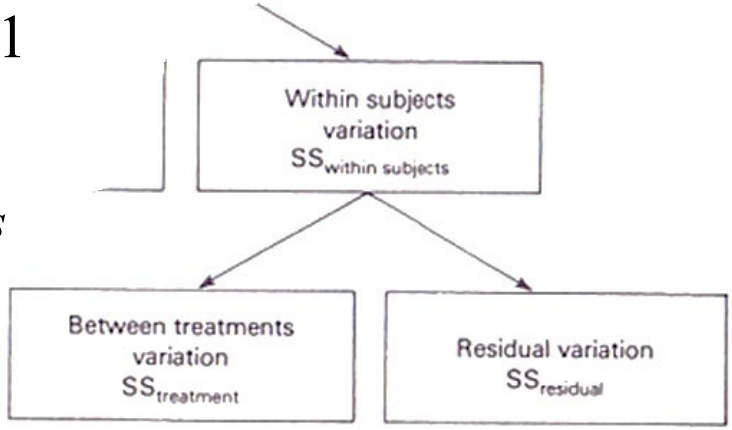
because each subject mean involves m treatments

Process: Separate Within Subject Sum of Squares into Treatment and Residual Sum of Squares

7. Sum of Squares Due to Treatment:

$$SS_{treat} = n\sum_{t=1}^{m}\left(\bar{T}_t - \bar{X}\right)^2 \qquad DF_{treat} = m-1$$

because each treatment mean involves n subjects



8. Sum of Squares Residual:

$$SS_{wit\,subjs} = SS_{treat} + SS_{res}$$

$$SS_{res} = SS_{wit\,subjs} - SS_{treat} \qquad DF_{res} = (n-1)(m-1)$$

9. Calculate *F* statistic:

$$F = \frac{SS_{treat} / DF_{treat}}{SS_{res} / DF_{res}} = \frac{MS_{treat}}{MS_{res}}$$

$$F_{crit} = F_{\alpha}(DF_{treat}, DF_{res})$$

$$DF_{treat} = m - 1 = \nu_n$$

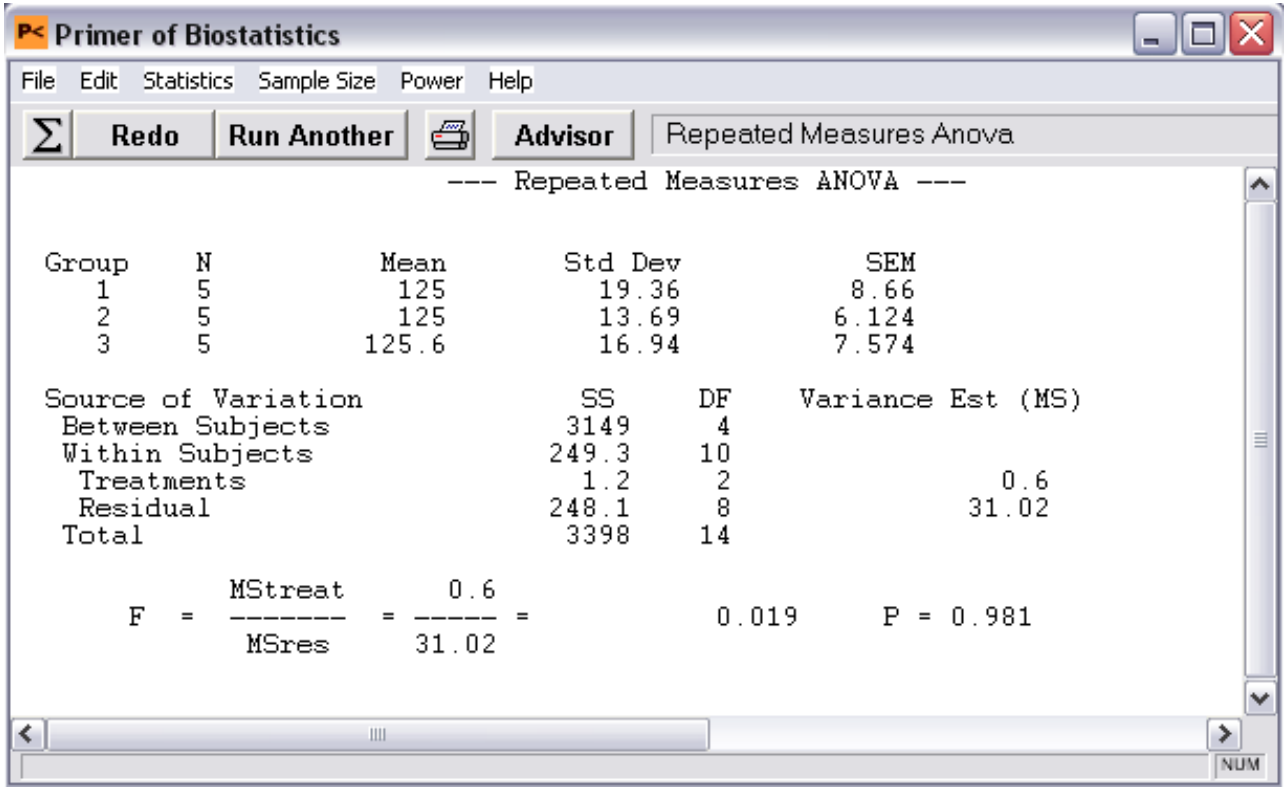
$$DF_{res} = (n-1)\cdot(m-1) = \nu_d$$

Example 1: Repeated Measures ANOVA

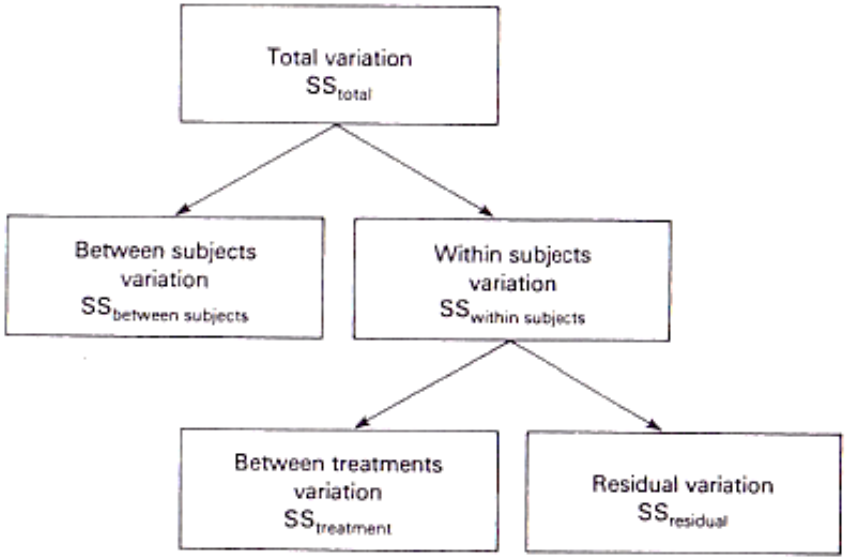
In a clinical trial, systolic blood pressures are measured in patients at 4, 8 and 12 weeks after administration of a vasodilator. Is there a difference in systolic blood pressures over time?

Systolic Blood Pressure (mmHg)

Subject	4 weeks	8 weeks	12 weeks
1	120	125	130
2	110	115	118
3	105	110	100
4	140	130	140
5	150	145	140



Example 1: Repeated Measures ANOVA



$$F = \frac{SS_{treat} / DF_{treat}}{SS_{res} / DF_{res}} = \frac{MS_{treat}}{MS_{res}}$$

Primer of Biostatistics

File Edit Statistics Sample Size Power Help

Redo | Run Another | Advisor | Repeated Measures Anova

--- Repeated Measures ANOVA ---

Group	N	Mean	Std Dev	SEM
1	5	125	19.36	8.66
2	5	125	13.69	6.124
3	5	125.6	16.94	7.574

Source of Variation	SS	DF	Variance Est (MS)
Between Subjects	3149	4	
Within Subjects	249.3	10	
Treatments	1.2	2	0.6
Residual	248.1	8	31.02
Total	3398	14	

$$F = \frac{MStreat}{MSres} = \frac{0.6}{31.02} = 0.019 \qquad P = 0.981$$

NUM

Example 2: Repeated Measures ANOVA

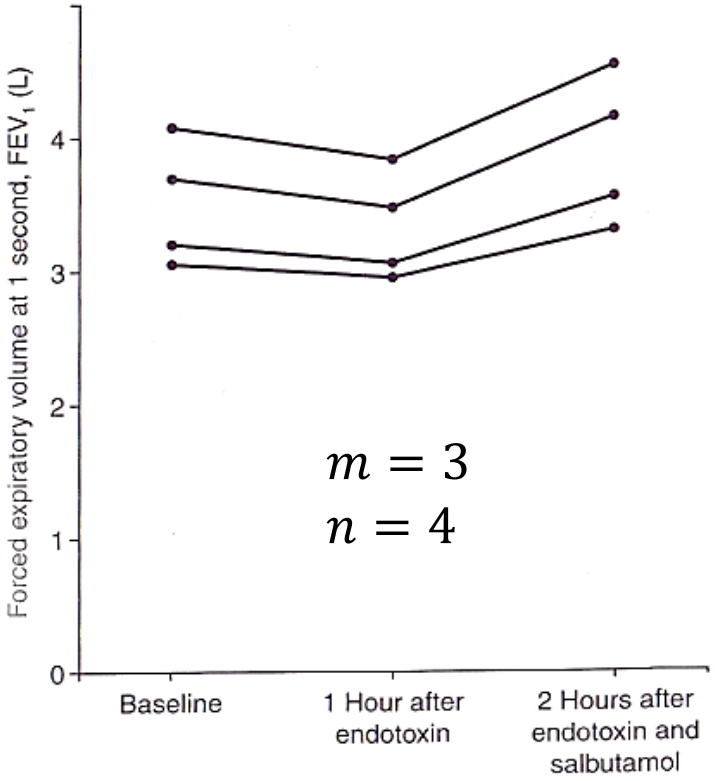


Table 9-5 Forced Expiratory Volume (L) at One Second before and after Bronchial Challenge with Endotoxin and Salbutamol Treatment

Person (subject)	No drug (baseline)	1 Hour after endotoxin	2 Hours after endotoxin and salbutamol	Subject	
				Mean	SS
1	3.7	3.4	4.0	3.70	0.1800
2	4.0	3.7	4.4	4.03	0.2467
3	3.0	2.8	3.2	3.00	0.0800
4	3.2	2.9	3.4	3.17	0.1267
Treatment mean	3.48	3.20	3.75		
	Grand mean = 3.48				SS_{tot} = 2.6830

Figure 9-6 Forced expiratory volume at one second (FEV_1) in four people at baseline, 1 hour after inhaling endotoxin, and 2 hours following endotoxin and salbutamol exposure. Each individual's response is connected by straight lines. *(Adapted from table 2 and fig. 4 of O. Michel, J. Olbrecht, D. Moulard, R. Sergysels, "Effect of Anti-asthmatic Drugs on the Response to Inhaled Endotoxin,"* Ann. Allergy Asthma Immunol. **85**: 305–310, 2000.)

H_0: There is no difference in FEV_1 under the 3 exposure conditions

$$SS_{wit\,subjs} = \sum_{s=1}^{4}\left(\sum_{t=1}^{3}\left(x_{ts} - \bar{S}_s\right)^2\right) = 0.6333 \qquad DF_{wit\,subjs} = n(m-1) = 8$$

$$SS_{treat} = n\sum_{t=1}^{m}\left(\bar{T}_t - \bar{X}\right)^2 = 0.6050 \qquad DF_{treat} = m-1 = 2$$

$$SS_{res} = SS_{wit\,subjs} - SS_{treat} = 0.0283 \qquad DF_{res} = (n-1)(m-1) = 6$$

$$F = \frac{MS_{treat}}{MS_{res}} = \frac{SS_{treat}/DF_{treat}}{SS_{res}/DF_{res}} = \frac{0.6050/2}{0.0283/6} = \frac{0.3025}{0.0047} = 64.053 \qquad P < 0.01$$

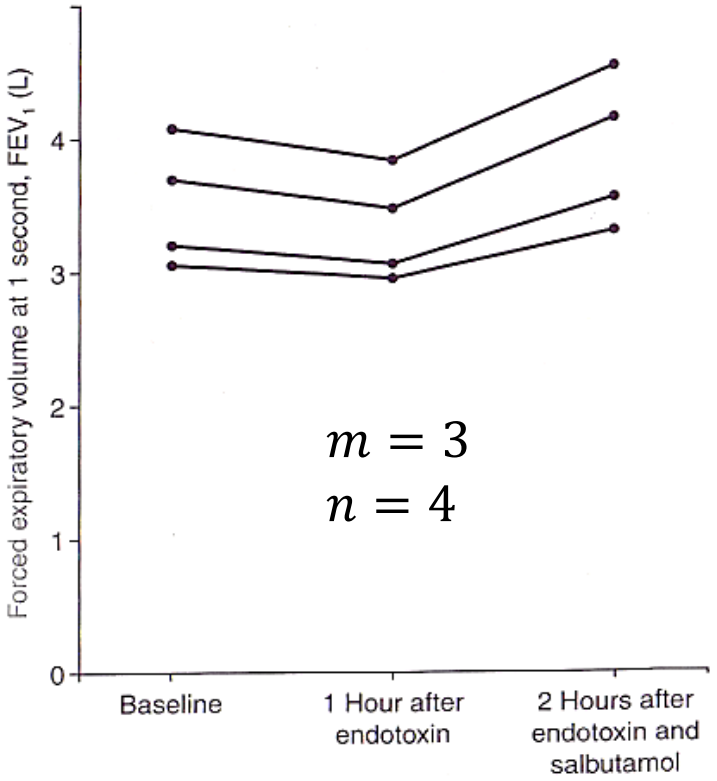
Table 9-6 Analysis-of-Variance Table for One-Way Repeated-Measures Analysis of FEV_1 in Endotoxin Response

Source of variation	SS	DF	MS
Between subjects	2.0490	3	
Within subjects	0.6333	8	
Treatments	0.6050	2	0.3025
Residual	0.0283	6	0.0047
Total	2.6830	11	

$$F = \frac{MS_{treat}}{MS_{res}} = \frac{0.3025}{0.0047} = 64.053$$

III. Isolating Differences in Repeated Measures ANOVA

Example 2 (continued)



$m = 3$

$n = 4$

Table 9-5 Forced Expiratory Volume (L) at One Second before and after Bronchial Challenge with Endotoxin and Salbutamol Treatment

Person (subject)	No drug (baseline)	1 Hour after endotoxin	2 Hours after endotoxin and salbutamol	Subject	
				Mean	SS
1	3.7	3.4	4.0	3.70	0.1800
2	4.0	3.7	4.4	4.03	0.2467
3	3.0	2.8	3.2	3.00	0.0800
4	3.2	2.9	3.4	3.17	0.1267
Treatment mean	3.48	3.20	3.75		
	Grand mean = 3.48				$SS_{tot} = 2.6830$

From previous results, conclude there is a difference among the 3 "treatments"/observation scenarios

$$F = 64.053 \quad P < 0.01$$

$$SS_{res} = SS_{wit\,subjs} - SS_{treat} = 0.0283$$

$$DF_{res} = (n-1)(m-1) = 6$$

$$MS_{res} = \frac{SS_{res}}{DF_{res}} = \frac{0.0283}{6} = 0.0047$$

Isolating Differences in Repeated Measures ANOVA

Holm-corrected multiple-comparison t Tests

$$t = \frac{\bar{T}_i - \bar{T}_j}{\sqrt{2MS_{res}/n}} \quad \text{for all } i,\ j \quad i \neq j$$

Use DF_{res} for t table DF

$$DF_{res} = (n-1)(m-1)$$

Order by |t|

		Significance Level (Holm correction)
Endotoxin vs. Salbutamol & Endotoxin	$t = \frac{3.20 - 3.75}{\sqrt{2(0.0047)/4}} = -11.346$	$0.05/3 = 0.017$
Baseline vs. Endotoxin	$t = \frac{3.48 - 3.20}{\sqrt{2(0.0047)/4}} = 5.776$	$0.05/2 = 0.025$
Baseline vs. Salbutamol & Endotoxin	$t = \frac{3.48 - 3.75}{\sqrt{2(0.0047)/4}} = -5.570$	$0.05/1 = 0.05$

---

significance to use for the jth pairwise comparison

$j = 1, \rightleftharpoons, k$

$$\alpha_j = \frac{\alpha_T}{(k - j + 1)}$$

α_T: Overall level of significance (e.g., 0.05)

k: # of pairwise comparisons

Summary of Some Statistical Methods to Test Hypotheses

Scale of measurement	Type of experiment: Two treatment groups consisting of different individuals	Three or more treatment groups consisting of different individuals	Before and after a single treatment in the same individuals	Multiple treatments in the same individuals	Association between two variables
Interval (and drawn from normally distributed populations*)	Unpaired *t* test (Chapter 4)	Analysis of variance (Chapter 3)	Paired *t* test (Chapter 9)	Repeated-measures analysis of variance (Chapter 9)	Linear regression, Pearson product-moment correlation, or Bland-Altman analysis (Chapter 8)
Nominal	Chi-square analysis-of-contingency table (Chapter 5)	Chi-square analysis-of-contingency table (Chapter 5)	McNemar's test (Chapter 9)	Cochrane Q†	Relative risk or odds ratio (Chapter 5)
Ordinal‡	Mann-Whitney rank-sum test (Chapter 10)	Kruskal-Wallis statistic (Chapter 10)	Wilcoxon signed-rank test (Chapter 10)	Friedman statistic (Chapter 10)	Spearman rank correlation (Chapter 8)
Survival time	Log-rank test or Gehan's test (Chapter 11)				

z Test

Logistic regression (slides)

*If the assumption of normally distributed populations is not met, rank the observations and use the methods for data measured on an ordinal scale.

†Not covered in this text.

‡Or interval data that are not necessarily normally distributed.

IV. Repeated Measures w/ Nominal Data – McNemar's Test

Presence of p7 antigen in breast cancer tumor cells before and after treatment with chemotherapy

		After	
		# Positive	# Negative
Before	# Positive	4	0
	# Negative	12	14

if we treated measurements as though they were all independent:

$$\chi^2 = \sum_{i=1}^{n_r}\sum_{j=1}^{n_c} \frac{\left(\left|O_{ij} - E_{ij}\right| - \frac{1}{2}\right)^2}{E_{ij}} = 2.165 \qquad v = \left(n_r - 1\right)\left(n_c - 1\right) = 1$$

(Yates corrected, because 2x2)

$$\chi^2 = 2.165 \not> \chi^2_{0.05}(1) = 3.842$$

Conclude no significant difference in p7 antigen expression before and after treatment

Presence of p7 antigen in breast cancer tumor cells before and after treatment

		After	
		# Positive	# Negative
Before	# Positive	4	0
	# Negative	12	14

Problem: Rows and columns are NOT independent
Experiment design is analogous to the paired t Test design

Solution: Focus on women that CHANGE status

If treatment has no effect, then would expect in the 12 women whose status changed:
½*12 (or 6) women who were positive before would be negative after
and
½*12 (or 6) women who were negative before would be positive after

$$\chi^2 = \frac{\left(|0-6|-1/2\right)^2}{6} + \frac{\left(|12-6|-1/2\right)^2}{6} = 10.083 \qquad \nu = 1$$

$$\chi^2 = 10.083 > \chi^2_{0.005}(1) = 7.879$$

Significant difference in p7 antigen expression before and after treatment
$P < 0.005$

McNemar's Test

		After	
		Outcome 1	Outcome 2
Before	Outcome 1	a	b
	Outcome 2	c	d

H_0: No effect of treatment; probability of change from Outcome 1 to Outcome 2 is same as probability of change from Outcome 2 to Outcome 1

Expected number to change under H_0

$$E = \frac{b+c}{2}$$

$$\chi^2 = \frac{\left(|b-E|-1/2\right)^2}{E} + \frac{\left(|c-E|-1/2\right)^2}{E} \qquad \nu = 1$$

(2 x 2 contingency table → use Yates's correction)

McNemar's Test

Used when:

- Testing individual response to treatment (e.g., previous example)
- Comparing results of two diagnostic procedures (see below)

Comparison of hypertension status as judged by an automated BP machine and an experienced person

	Person	
BP Machine	+	−
+	3	7
−	1	9

according to the gold standard method, none of these 20 subjects has hypertension

H_0: There is no difference in specificity between the two methods of evaluating hypertension (specificity: the ability of a test to correctly identify true negatives)

$$E = \frac{1+7}{2} = 4 \qquad \chi^2 = \frac{\left(|7-4|-1/2\right)^2}{4} + \frac{\left(|1-4|-1/2\right)^2}{4} = 3.125 \qquad \nu = 1$$

$$\chi^2 = 3.125 \not> \chi^2_{0.05}(1) = 3.841$$